

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 Real Analysis Lecture 12
 - Subsequences
 - Cauchy sequences

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Existence of limits

Recall a **subsequence** of a sequence $\{x_n\}$ is a sequence of the form $\{x_{k(n)}\}$, for $k(1) < k(2) < \dots$ positive integers.

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Theorem

A sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$ if and only if every subsequence $\{x_{k(n)}\}$ also converges to L .

Existence of limits

Proof.

Suppose that every subsequence $\{x_{k(n)}\}$ of $\{x_n\}$ converges to L .

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Let $\{x_{k(n)}\}$ be a subsequence of $\{x_n\}$.

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To show it limits to L , let $\epsilon > 0$.

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Let $\{x_{k(n)}\}$ be a subsequence of $\{x_n\}$.

To show it limits to L , let $\epsilon > 0$.

Then there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d(x_n, L) < \epsilon$.

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Let $\{x_{k(n)}\}$ be a subsequence of $\{x_n\}$.

To show it limits to L , let $\epsilon > 0$.

Then there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d(x_n, L) < \epsilon$.

Moreover, $k(n) \geq n$ so $k(n) \geq N$ and therefore $d(x_{k(n)}, L) < \epsilon$.

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Moreover, $k(n) \geq n$ so $k(n) \geq N$ and therefore $d(x_{k(n)}, L) < \epsilon$.

Since $\epsilon > 0$ was arbitrary, this proves convergence. $\square$

Monotone sequences

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In any of these cases, we call the sequence **monotone**.

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A monotone increasing sequence $\{x_n\}$ of real numbers which is bounded above converges to $L = \sup\{x_1, x_2, \dots\}$.

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Theorem (Monotone convergence theorem)

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A monotone increasing sequence $\{x_n\}$ of real numbers which is bounded above converges to $L = \sup\{x_1, x_2, \dots\}$.

A monotone decreasing sequence $\{x_n\}$ which is bounded below converges to $L = \inf\{x_1, x_2, \dots\}$.

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Suppose that $\{x_n\}$ is a monotone increasing sequence which is bounded above and let $X = \{x_1, x_2, \dots\}$.

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Let $\epsilon > 0$.

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Proof.

Suppose that $\{x_n\}$ is a monotone increasing sequence which is bounded above and let $X = \{x_1, x_2, \dots\}$. Then by the Completeness Axiom $L = \sup X$ exists.

Let $\epsilon > 0$.

Then $L - \epsilon$ is not a supremum of X , so there exists $x_N \in X$ such that $x_N > L - \epsilon$.

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Then $L - \epsilon$ is not a supremum of X , so there exists $x_N \in X$ such that $x_N > L - \epsilon$.

Since the sequence is increasing, for any $n > N$, we have that $L - \epsilon < x_N \leq x_n < L$.

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Since the sequence is increasing, for any $n > N$, we have that $L - \epsilon < x_N \leq x_n < L$.

It follows that

$$d(x_n, L) = |x_n - L| = L - x_n < \epsilon.$$

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Since $\epsilon > 0$ was arbitrary, this proves x_n converges to L . □

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$\forall \epsilon > 0$, there exists $N \in \mathbb{Z}_+$ with $m, n \geq N \Rightarrow d(x_m, x_n) < \epsilon$.

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- is every Cauchy sequence convergent?

Challenge!

Problem

Prove that if $\{x_n\}$ is a Cauchy sequence in a metric space (M, d) , then it is bounded.

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Suppose $\{x_n\}$ is Cauchy.
Let $\epsilon = 2025$.

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Suppose $\{x_n\}$ is Cauchy.

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Then $\exists N \in \mathbb{Z}_+$ with $d(x_m, x_n) < 2025$ for all $m, n \geq N$.

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Then $\exists N \in \mathbb{Z}_+$ with $d(x_m, x_n) < 2025$ for all $m, n \geq N$.

Therefore for all $n \geq N$, we have $d(x_n, x_N) \leq 2025$.

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So if we define

$$R = \max\{d(x_1, x_N) + 1, d(x_2, x_N) + 1, \dots, d(x_{n-1}, x_N) + 1, 2025\},$$

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then $d(x_n, x_N) \leq 2025$ for all $n \geq 1$.

Hence $X = \{x_1, x_2, \dots\} \subseteq B_M(x_N, R)$, and is bounded.

Cauchy sequence

Every convergent sequence is Cauchy.

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Theorem

Let (M, d) be a metric space and suppose that $\{x_n\}$ is a sequence in M which converges to $L \in M$.

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Let (M, d) be a metric space and suppose that $\{x_n\}$ is a sequence in M which converges to $L \in M$.

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Therefore for any $m, n \geq N$ we have

$$d(x_m, x_n) \leq d(x_m, L) + d(L, x_n) < \epsilon/2 + \epsilon/2 = \epsilon.$$

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Partial converse.

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Let $M = \mathbb{R}^n$ with the Euclidean metric, and suppose that $\{x_n\}$ is a Cauchy sequence in M .

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- when X is finite

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We consider two cases:

- when X is finite
- when X is infinite



Cauchy sequence

Proof.

Case I: Assume X is finite.

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Then

$$\{d(x, y) : x, y \in X, x \neq y\}$$

is a finite set of positive values.

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Since $\{x_n\}$ is Cauchy, there exists $N \in \mathbb{Z}_+$ with $d(x_m, x_n) < \epsilon$ for all $m, n \geq N$.

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Since $\{x_n\}$ is Cauchy, there exists $N \in \mathbb{Z}_+$ with $d(x_m, x_n) < \epsilon$ for all $m, n \geq N$.

This implies $x_n = x_N$ for all $n \geq N$.

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Hence $\lim_{n \rightarrow \infty} x_n = x_N$. □

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Then by the Bolzano-Weierstrass Theorem, X has an accumulation point $L \in M$.

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Let $\epsilon > 0$.

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Since $\{x_n\}$ is Cauchy, we can choose $N \in \mathbb{Z}_+$ with $d(x_m, x_n) < \epsilon/2$ for all $m, n \geq N$.

Moreover, the ball $B(L, \epsilon/2)$ contains infinitely many points of X , so we can choose $\ell \geq N$ with $x_\ell \in B(L, \epsilon/2)$.

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- *tries* to converge to $\sqrt{2}$
- however $\sqrt{2} \notin M$, so it doesn't converge!
- somehow M is "missing" some points

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- $[0, 1]$ with the Euclidean metric is complete

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- $\mathbb{R}^n$ with the Euclidean metric is complete
- $[0, 1]$ with the Euclidean metric is complete
- $(0, 1)$ with the Euclidean metric is not complete

Challenge!

Problem

Show that the interval $(0, 1) \subseteq \mathbb{R}$ with the Euclidean metric is not a complete metric space.

Complete spaces

Theorem

Let (M, d) be a metric space.

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Let (M, d) be a metric space. Then for any compact subset $S \subseteq M$, the subspace (S, d) is complete.

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Proof.

Let (M, d) be a metric space and $S \subseteq M$ compact.

Complete spaces

Theorem

Let (M, d) be a metric space. Then for any compact subset $S \subseteq M$, the subspace (S, d) is complete.

Proof.

Let (M, d) be a metric space and $S \subseteq M$ compact.
To show that (S, d) is complete, we must show that every Cauchy sequence in S converges to a value in S .

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TIME OUT!



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Then for all $x \in S$, there exists $r_x > 0$ such that $B_S(x, r_x) \cap X$ is empty if $x \notin X$ or equal to $\{x\}$ if $x \in X$.

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The open cover $\{U_i : i \in I\}$ with $I = S$ and $U_i = B_S(i; r_i)$ of S has no finite subcover.

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This is a contradiction, so S has an accumulation point of X . □

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Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to L . □